

DAY 1: DIFFERENTIATION - NOTES

RATES OF CHANGE IN MECHANICS

Goal

Achieve fluency in ALL differentiation techniques (power, product, quotient, chain, implicit, parametric, logarithmic, higher orders) by deriving every kinematics quantity possible. By the end of today, you will be able to take the derivative of any function and apply it to motion in one and two dimensions, as well as optimize physical quantities.

1 THE DERIVATIVE - DEFINITION AND INTERPRETATION

1.1 Definition of the Derivative

The derivative of a function $f(x)$ at a point x is defined as:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Interpretation:

- **Instantaneous rate of change** of f with respect to x
- **Slope of the tangent line** to the graph of f at the point $(x, f(x))$

2 DIFFERENTIATION RULES

2.1 Power Rule

$$\boxed{\frac{d}{dx}[x^n] = nx^{n-1}}$$

This holds for all real numbers n (positive, negative, fractional).

Examples:

$$\begin{aligned}\frac{d}{dx}[x^5] &= 5x^4 \\ \frac{d}{dx}[x^{-3}] &= -3x^{-4} \\ \frac{d}{dx}[x^{1/2}] &= \frac{1}{2}x^{-1/2}\end{aligned}$$

2.2 Constant Multiple Rule

$$\boxed{\frac{d}{dx}[c \cdot f(x)] = c \cdot f'(x)}$$

2.3 Sum/Difference Rule

$$\boxed{\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)}$$

2.4 Product Rule

$$\boxed{\frac{d}{dx}[u(x)v(x)] = u'v + uv'}$$

Mnemonic: "First times derivative of second plus second times derivative of first."

2.5 Quotient Rule

$$\boxed{\frac{d}{dx} \left[\frac{u(x)}{v(x)} \right] = \frac{u'v - uv'}{v^2}}$$

Mnemonic: "Lo dee Hi minus Hi dee Lo, over Lo Lo."

2.6 Chain Rule (The Most Important Rule)

$$\boxed{\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)}$$

In Leibniz notation: if $y = f(u)$ and $u = g(x)$, then:

$$\boxed{\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}}$$

Interpretation: Derivative of outside (evaluated at inside) times derivative of inside.

2.7 Implicit Differentiation

Use when y is not explicitly defined as a function of x (e.g., equations like $x^2 + y^2 = 25$).

Procedure:

1. Differentiate both sides of the equation with respect to x , treating y as $y(x)$
2. Apply the chain rule: $\frac{d}{dx}[y^n] = ny^{n-1}\frac{dy}{dx}$
3. Collect all $\frac{dy}{dx}$ terms on one side
4. Solve for $\frac{dy}{dx}$

2.8 Parametric Differentiation

If $x = f(t)$ and $y = g(t)$, then:

$$\boxed{\frac{dy}{dx} = \frac{dy/dt}{dx/dt}}$$

provided $dx/dt \neq 0$.

2.9 Logarithmic Differentiation

Useful for complicated products, quotients, and powers.

Procedure:

1. Take natural logarithm of both sides: $\ln y = \ln[f(x)]$
2. Simplify using logarithm properties
3. Differentiate implicitly with respect to x
4. Solve for y'

2.10 Higher Order Derivatives

$$\text{First derivative: } f'(x) = \frac{df}{dx}$$

$$\text{Second derivative: } f''(x) = \frac{d^2 f}{dx^2}$$

$$\text{Third derivative: } f'''(x) = \frac{d^3 f}{dx^3}$$

$$\text{nth derivative: } f^{(n)}(x) = \frac{d^n f}{dx^n}$$

3 DERIVATIVES OF SPECIAL FUNCTIONS

3.1 Trigonometric Functions

$$\begin{aligned}\frac{d}{dx}[\sin x] &= \cos x \\ \frac{d}{dx}[\cos x] &= -\sin x \\ \frac{d}{dx}[\tan x] &= \sec^2 x \\ \frac{d}{dx}[\cot x] &= -\csc^2 x \\ \frac{d}{dx}[\sec x] &= \sec x \tan x \\ \frac{d}{dx}[\csc x] &= -\csc x \cot x\end{aligned}$$

3.2 Exponential and Logarithmic Functions

$$\begin{aligned}\frac{d}{dx}[e^x] &= e^x \\ \frac{d}{dx}[a^x] &= a^x \ln a \\ \frac{d}{dx}[\ln x] &= \frac{1}{x} \\ \frac{d}{dx}[\log_a x] &= \frac{1}{x \ln a}\end{aligned}$$

4 PHYSICS INTERPRETATIONS

4.1 Kinematics

For a particle moving along a line with position $x(t)$:

$$\begin{aligned}\text{Velocity: } v(t) &= \frac{dx}{dt} \\ \text{Acceleration: } a(t) &= \frac{dv}{dt} = \frac{d^2x}{dt^2} \\ \text{Jerk: } j(t) &= \frac{da}{dt} = \frac{d^3x}{dt^3}\end{aligned}$$

4.2 Related Rates

When quantities are related by an equation, their rates of change are related by differentiating with respect to time.

4.3 Parametric Motion in 2D

For a particle with position $(x(t), y(t))$:

$$\text{Velocity components: } v_x = \frac{dx}{dt}, \quad v_y = \frac{dy}{dt}$$

$$\text{Speed: } |\vec{v}| = \sqrt{v_x^2 + v_y^2}$$

$$\text{Direction: } \theta = \tan^{-1} \left(\frac{v_y}{v_x} \right)$$

$$\text{Acceleration components: } a_x = \frac{dv_x}{dt}, \quad a_y = \frac{dv_y}{dt}$$

4.4 Tangential and Normal Acceleration

For a particle moving along a curved path:

$$\text{Tangential acceleration: } a_t = \frac{d|\vec{v}|}{dt}$$

$$\text{Normal acceleration: } a_n = \frac{v^2}{\rho}$$

where ρ is the radius of curvature.

For parametric curves, the radius of curvature is:

$$\rho = \frac{(\dot{x}^2 + \dot{y}^2)^{3/2}}{|\dot{x}\ddot{y} - \dot{y}\ddot{x}|}$$

5 OPTIMIZATION - FINDING MAXIMA AND MINIMA

5.1 Theory

One of the most powerful applications of derivatives in physics is finding the maximum or minimum value of a physical quantity. This could be:

- Maximum height of a projectile
- Minimum time for light to travel (Fermat's principle)
- Maximum range of a projectile
- Minimum energy configuration
- Maximum speed in a given motion

5.2 Procedure for Optimization

1. Identify the quantity Q to be optimized (maximized or minimized).
2. Express Q as a function of a single variable x : $Q = f(x)$.
3. Find critical points by solving $f'(x) = 0$.
4. Use the first or second derivative test to determine if each critical point is a maximum, minimum, or neither.
5. Check endpoints if the domain is restricted.
6. Interpret the result in the physics context.

5.3 First Derivative Test

- If f' changes from $+$ to $-$ at c , then $f(c)$ is a local maximum.
- If f' changes from $-$ to $+$ at c , then $f(c)$ is a local minimum.
- If f' does not change sign, then $f(c)$ is neither (inflection point).

5.4 Second Derivative Test

- If $f''(c) < 0$, then $f(c)$ is a local maximum.
- If $f''(c) > 0$, then $f(c)$ is a local minimum.
- If $f''(c) = 0$, the test is inconclusive.

5.5 Key Optimization Problems in Physics

- **Projectile motion:** Maximum height (when $v_y = 0$), maximum range (when $\theta = 45^\circ$)
- **Geometric optimization:** Minimum surface area for a given volume (often gives $h = 2r$ for cylinders)
- **Fermat's principle:** Light takes path of least time $\rightarrow$ Snell's Law: $\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}$
- **Electrical circuits:** Maximum power transfer when $R_{\text{load}} = R_{\text{internal}}$
- **River crossing:** Minimum time vs. reaching a specific point

Derivative Rules Cheat Sheet

- **Power Rule:** $\frac{d}{dx}[x^n] = nx^{n-1}$
- **Product Rule:** $\frac{d}{dx}[uv] = u'v + uv'$
- **Quotient Rule:** $\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{u'v - uv'}{v^2}$
- **Chain Rule:** $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$
- **Implicit Differentiation:** Differentiate both sides with respect to x , treating y as $y(x)$, then solve for $\frac{dy}{dx}$
- **Parametric:** $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$
- **Logarithmic:** Take $\ln$ of both sides, differentiate implicitly, then solve for y'
- **Trig Derivatives:** $\frac{d}{dx}[\sin x] = \cos x$, $\frac{d}{dx}[\cos x] = -\sin x$, $\frac{d}{dx}[\tan x] = \sec^2 x$
- **Exponential/Log:** $\frac{d}{dx}[e^x] = e^x$, $\frac{d}{dx}[\ln x] = \frac{1}{x}$

Optimization Tests

- **First Derivative Test:**
 - f' changes $+$ $\rightarrow$ $-$: local maximum
 - f' changes $-$ $\rightarrow$ $+$: local minimum
 - f' doesn't change sign: neither
- **Second Derivative Test:**
 - $f''(c) < 0$: local maximum
 - $f''(c) > 0$: local minimum
 - $f''(c) = 0$: test inconclusive